RossijskijFeatrDB: Modeling Russian Imperial Opera Sources as Linked Data

Anna Giust[1], Martina Pernigotto[1] and Matteo Lissandrini[1]

[1]Department of Foreign Languages and Literatures, University of Verona, Verona, Italy

Abstract

Musicology aims to reconstruct historical landscapes by treating music as a living, multiethnic cultural system of practices. When it comes to analyzing the specific context of opera, we require not only cataloguing diverse primary sources, such as (hand- and typewritten) librettos, scores, playbills, and press accounts, but also building a systematic and reliable record of actual theatrical practices, establishing and representing correct relationships among heterogeneous sources. Here, we apply this approach to the *Rossiysky Θeatr* project, which aims to recover the continuous, multilingual operatic life of the Russian Empire in the years 1730 to 1836 by creating an open-access digital archive of its stages and repertories. Building such an archive is highly challenging, as it requires organizing a vast, fragmented corpus of primary documents in multiple languages while modeling the complex, fluid relationships between librettos, their translations, music composed to mirror these changing verbal texts and other circumstantial elements. Currently, the project is gathering data on over 600 operas, documented through thousands of digital scans of printed sources and manuscripts that are otherwise inaccessible to musicologists. To model these historical records with modern data standards, we designed a web-based curation application for data entry and digital cataloguing built on semantic web technologies like SHACL. By aligning our system with LRMoo, CIDOC-CRM, and the Polifonia ontologies (Core and Music Meta), we show how to dynamically generate form-based editing, validation, and autocomplete features directly from the underlying RDF schema. This schema-driven, low-code architecture allows the application to adapt instantly as our understanding of these complex performing arts relationships evolves.

Keywords

Semantic Web, Knowledge Graphs, Linked Data, Music Sources, Digital Humanities, RDF

1. Introduction

Rossiysky Θeatr (https://rosfeatr.eu/), is a project based at the University of Verona investigating operatic life across the historical territories of the Russian Empire. The project rejects teleological and nationalizing accounts of music history, documenting instead a highly multilingual, fluid theatrical landscape in which Italian, French, German, Polish, Ukrainian, Russian, and other traditions shared stages, personnel, repertoire, and practices, which left diverse documentary traces. Systematically cataloguing these traces is not merely an archival task, but a critical prerequisite for a deeper musicological understanding of how diverse cultural identities intersected in performance.

However, modeling the performing arts presents profound conceptual challenges [1]. Unlike static library materials, a live performance is ephemeral and immaterial. As a result, traditional cataloguing frequently conflates the live event with its static "pre-text" (e.g., a written score or script). Standard conceptual reference models like the IFLA Library Reference Model (LRM) and its object-oriented formulation LRMoo [2], are historically designed for material, bibliographic artifacts and fail to cleanly capture the ontology of live performances. Determining what constitutes the core "work of art" is notoriously difficult in the performing arts [1]: it involves disentangling the original composition from its translation, its specific production, and the complex, multi-layered web of historic agents, librettos, scores, and local adaptations in changing geographic and cultural contexts that bring a staging to life. The case addressed by Rossiysky Θeatr is made even more complex due to the dispersion of sources and their dissemination in locations that challenge traditional approaches based on locations (specific funds, archives, or national approaches).

ISWC 2026 Companion Volume, October 25–29, 2026, Bari, Italy

anna.giust@univr.it (A. Giust); matteo.lissandrini@univr.it (M. Lissandrini)



© 2026 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

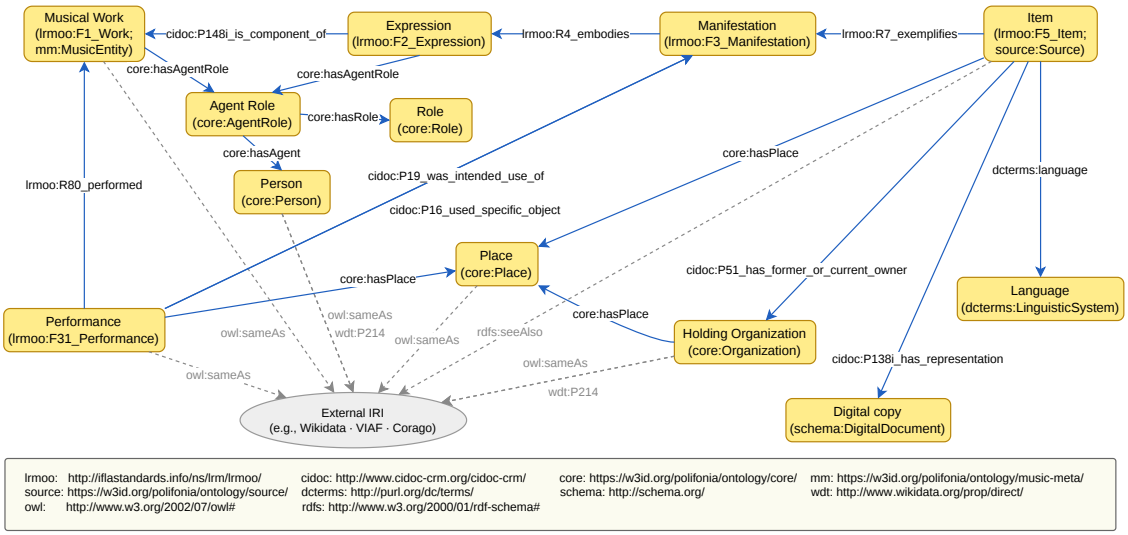


Figure 1: Simplified representation of the core data model for RossijskijFeatrDB

To address these modeling challenges, we present RossijskijFeatrDB, an RDF dataset and digital archive designed to represent theatrical works, libretti, performances, editions and manuscripts along with their source witnesses, agents, roles, places, and dates, and enriched with external authority links. Linked Data and Semantic Web technologies are uniquely suited to model this corpus because they allow us to represent these heterogeneous, highly interconnected relationships without forcing them into rigid, flat hierarchies. Yet, to allow curators to build RossijskijFeatrDB, we faced important modeling questions and the need for new data-management tools. On the modeling side, we decided to align our schema with modern ontological frameworks, and in particular LRMoo [2], CIDOC-CRM [3], and the Polifonia ontology [4].

Hence, as our first contribution, we show how semantic technologies can bridge the gap between material bibliography and the representation of immaterial performance. Furthermore, our main technical contribution is a standalone, schema-driven curation application [1] guided by SHACL Shapes [5]. This application dynamically generates user-facing data entry forms, real-time validations, and autocomplete fields directly from the underlying RDF schema. This approach provides domain curators with a low-code, highly adaptable environment to model, validate, and query the complex network of relationships between performances, artists, events, and documents. We believe this work acts also as a practical showcase of the advantages offered by employing validating shapes for data curation and a replicable model for similar efforts in other digital archive initiatives.

2. Modeling Approach

RossijskijFeatrDB adopts the FRBR/LRM Work/Expression/Manifestation/Item (WEMI) structure to model bibliographic and documentary evidence at the appropriate level of abstraction [6, 2] (see Figure 1): Work for the distinct intellectual or artistic creation, Expression for linguistic or intellectual realizations connected to it, Manifestation for a physical embodiment, and Item for an individual copy held by a library or archive (realized in our schema as a node typed jointly as `lrmoo:F5_Item` and Polifonia `source:Source` class). This distinction is essential for opera sources, because the same theatrical work may be documented through multiple libretto editions, translations, scores, parts, and surviving copies, each with different dates, places, languages, annotations, and holding institutions.

Modeling alternatives. Several existing frameworks offer partial solutions to representing performing-arts materials, each with trade-offs that motivated our specific combination of standards. Our analysis

[1] https://github.com/VIASystemsLab/rfdb-curator/

is based on recent analysis of how existing models have been applied to the concept of "work" [1]. The original FRBR entity-relationship model and its object-oriented successor FRBRoo [6] introduced the Work/Expression/Manifestation/Item (WEMI) hierarchy and, through harmonization with CIDOC-CRM [3], an event-centric treatment of creation. Unfortunately, FRBRoo's bibliographic orientation leaves performance events awkwardly attached to a model built primarily for texts. LRMoo [2], FRBRoo's successor, generalizes WEMI and introduces `F31_Performance` as a first-class class alongside Work. Yet, modeling the relationship between a performance and intermediate creative components such as a libretto or a score remains cumbersome in practice, typically requiring verbose combinations of generic LRMoo constructs [1]. A similar tension recurs across other CIDOC-CRM-based cultural-heritage projects. ArCo [7] extends CIDOC-CRM for tangible and intangible Italian cultural heritage but does not model bibliographic WEMI relations at all. Instead, IMAGO [8] extends CIDOC-CRM and LRMoo specifically for manuscript and textual traditions, which resembles our libretto and score materials, but does not face the challenges to model composite entities like operas. For the musical domain specifically, the Polifonia ontology network [4], and its Music Meta and Core modules in particular, offered a closer fit: Music Meta was explicitly designed to sit along LRMoo/CIDOC-CRM for musical works, while Core supplies a lightweight, reusable agent/role/place vocabulary.

Our modeling is strongly inspired by Corago [9], which demonstrated the applicability of CIDOC-CRM and FRBRoo to opera. Unlike Corago, and in contrast to fully materialized LRMoo implementations, however, we adopt a more pragmatic interpretation of LRMoo, simplifying parts of the WEMI structure to accommodate the composite nature of operas with a more succinct representation.

2.1. Summary of Design Decisions

RossiJskijFeatrDB's schema reuses the LRMoo and CIDOC-CRM ontologies and vocabularies but *with three deliberate departures from a straightforward application of their data model.* First, while we retain the WEMI hierarchy, we do not materialize the Work-level entities for the distinct components of an opera. In simplified terms, that means, for example, that we have the Expression level entity for a libretto and a score, but not their corresponding Work level entities. Hence, each Expression-level entity of librettos and scores points to their parent opera Work directly via `cidoc:P148i_is_component_of`, avoiding the proliferation of unmaterialized intermediate Work nodes that a strict WEMI decomposition of "the libretto" versus "the opera" versus "the score" would otherwise require. Second, we use the Polifonia Core vocabulary [4] to represent the connection between people and their artistic creation, e.g., the composer, the librettist, or the translator. This departs from the LRMoo/CIDOC-CRM event based modeling that is more verbose since it requires an intermediate event entity instead. Third, we treat `lrmoo:F31_Performance` as a first-class citizen linked to both the Work it performed (`lrmoo:R80_performed`) and, where relevant, the specific Manifestation used in staging it, distinguishing a strong claim of intended use (`cidoc:P19_was_intended_use_of`) from a weaker claim of mere presence (`cidoc:P16_used_specific_object`). The distinction is motivated by the fact that in most cases artifacts like librettos were printed on purpose for a given staging, while in some rare cases it is possible that librettos may have been re-used for subsequent replicas. These choices keep the schema anchored in established and interoperable standards while remaining pragmatic about the economy of the data model and the modeling expressivity required by musicologists.

Opera creates a specific modeling challenge because it is a composite artistic work produced through the joint contribution of composer and librettist, working across music and text. Unlike what is typical in popular music, in the operas considered here, the libretto frequently functions as the conceptual blueprint from which the composer works. Translating an opera, then, entails a transformation, as the music already exists, and changing the language disrupts the close relationship between words and music, the music has to be adapted to accommodate the translated text. In this model, the opera is therefore represented as a single LRMoo `F1 Work` identified by that joint contribution, not as a composer's musical work to which a text is merely attached. At the same time, the model preserves the independent documentary life of librettos and scores, since historical collections often hold printed libretto editions independently of any associated score.

The case of *La forza dell'amore e dell'odio*, with music by Francesco Araja and libretto by Francesco Prata, illustrates the problem. Productions of this opera are attested by different historical editions of its libretto, including a libretto printed in Milan in 1734 for the production of Regio Ducal Teatro di Milano (held as `I-Mb RACC.DRAM.6047 003`) and a bilingual Italian/German libretto printed in St. Petersburg for the staging in 1736 (held as `RUS-SPsc, Rossica 13.8.5.44`). The opera was performed in Italian both times, but the Italian text in the two librettos is different. This implies that in some parts also the score of the two performances had to be different. In line with the musicological practice in this case, the two versions of the Italian text along the German translation are all associated to the same `F1 Work` entity for the opera. Yet, each edition has its own WEMI profile for the rest of the hierarchy: a textual Expression, a printed Manifestation with a place, year, and language configuration, and an Item corresponding to a specific archival or library copy. A fully materialized WEMI graph would create an independent Libretto Work for Prata's text, connect the Opera Work to that Libretto Work with the appropriate work-level relation, and then connect the Libretto Work to its Expressions, Manifestations, and Items. We decided to avoid producing an independent Libretto Work node for which we have little use and no additional information compared to its expression level entity. Hence, pragmatically, the intermediate Libretto Work is left unmaterialized unless it becomes analytically necessary. The shortcut is ontologically safe because both `F1 Work` and `F2 Expression` descend from CIDOC-CRM `E89 Propositional Object`, so `P148i is component of` can connect them as component propositional objects.

3. RFDB Curator: SHACL-Driven Editorial Infrastructure

Our second main contribution is RFDB Curator, a standalone web application, released fully open source[1], that lets musicologists enter data in RossijskijFeatrDB. Its defining feature is that it parses the SHACL shapes describing the data model to *generate the required forms for data entry*: every `sh:NodeShape` becomes a record type and every `sh:property` becomes a form field, so when the model evolves, forms, validation, and autocomplete follow without touching application code. Notably, RFDB Curator does not require the shapes to contain any presentation-oriented attribute.

As a matter of fact, existing shape-driven editors require extending the shape graph with a presentation layer that curators have to write and maintain by hand. In particular, DASH [10] annotates each property shape with a `dash:editor` drawn from a catalogue of widget classes (e.g., `dash:AutoCompleteEditor`, `dash:DatePickerEditor`, `dash:DetailsEditor`), and the SHACL 1.2 UI draft [11] standardizes the same idea as `shui:editor`, `shui:viewer`, and `shui:propertyRole`. Such terms are non-validating by design: they make the schema double as the configuration file of one particular interface, they must be kept in step with the constraints they decorate, and no validator can report when the two drift apart. We adopt instead an application-specific *opinionated* mapping, in which the widget is a deterministic function of the constraints a curator has to satisfy anyway: `sh:in` yields a closed-choice selector, `rdf:langString` with no upper bound a repeatable language-tagged field, `xsd:date` or `xsd:gYear` a temporal input at the declared granularity, `sh:class` or `sh:node` an autocomplete search over existing records, and `sh:minCount`/`sh:maxCount` requiredness and repeatability. Moreover, the distinction between entities and relations is also derived from the shapes rather than declared. That is, where SHACL-UI expects an explicit `shui:LabelRole`, for RFDB Curator a shape that constrains `rdfs:label` is treated as a first-class record, searchable and editable on its own, while a shape without a label, in our case `core:AgentRole`, is considered a *bridge entity* working as a reification of a relation. For example, `core:AgentRole` reifies the relation between a Work, a Person, and a Role. In such cases, the information of a bridge entity is editable only inline, within the form of its parent. This approach, compared to other solutions, has the limitation of not allowing per-field presentational overrides, e.g., signaling that a prose field should present a textarea. On the other hand, our shapes carry no presentation triples at all, the schema requires only the statements a validator consumes, which means that any SHACL Core shape graph becomes editable without any preparatory annotation step. The division of labor between interface and validator is therefore explicit:

the form layer reads only the constraint components that restrict how a value may be entered, namely `sh:in`, `sh:datatype`, `sh:class`/`sh:node`, and `sh:minCount`/`sh:maxCount` for widget selection, plus `sh:pattern`, which it compiles into client-side input validation. Purely assertional components, such as `sh:closed` and `sh:hasValue`, are never interpreted by the interface and are enforced by the validator alone, so no constraint needs a presentational counterpart for a shape graph to be usable. In our current implementation, during data entry, records are submitted as JSON-LD, validated on the server with pySHACL [12] against the active schema, and persisted to an Oxigraph [13] store via SPARQL and the Graph Store Protocol; the complete runtime ships via Docker Compose.

Our second core feature addresses validation during *incremental* editing, where curators add one entity at a time. For example, a new Expression may refer to an existing Work that already carries helper nodes, e.g., the aforementioned `core:AgentRole` bridge nodes; validating such a payload in isolation would raise spurious SHACL violations. To prevent this, the backend validates each submission against a merged graph that pulls in the referenced entities already stored. This merge is transitive and bounded by the connectivity of the shape graph, so all relevant helper nodes are included automatically and curators never resubmit the surrounding graph. Our implementation of this mechanism is deliberately simple. At start-up, the application compiles a shape dependency graph. This graph is used to build merged graph issuing a CONSTRUCT query retrieving the context connected to the given entity edited. Hence, this merged graph is compiled per request and then re-validated in full. This is adequate for single-record editorial writes which we do not expect would span huge portions of the graph. Yet, it does more work than necessary, since it re-checks every shape reachable from the edited record rather than only those a given change can actually violate. SHACL validation under graph updates has been characterized formally [14] and the issues connected to it inform our current and future work.

In particular, recent work provides principled remedies to the issue of scalable SHACL validation under specific node updates. In particular, our work is inspired by UpSHACL [15] that identifies the subgraph affected by an update and validates a reduced subgraph with an off-the-shelf engine and by Trav-SHACL [16], which propose to plan the traversal of a shape network so that invalid entities are detected early. In the future, we plan to adopt such targeted validation mechanisms to ensure RFDB Curator can scale-up to a growing collection.

4. Conclusions and Future Work

We presented RossijskijFeatrDB, an effort to model the multilingual operatic life of the Russian Empire as Linked Data. Our contribution is twofold: (1) a pragmatic alignment of LRMoo, CIDOC-CRM, and the Polifonia ontologies that captures composite theatrical evidence; and (2) RFDB Curator, a standalone application that generates its entire editing interface from validating SHACL shapes. Thanks to this, we are helping musicologists publish and grow an online digital archive of over 600 operas and their digitised sources, which we will make fully available online. On the modeling and tooling side, we will continue growing the reusable SHACL shapes and the curation application as a replicable resource for comparable digital-archive initiatives, and to evaluate the model against concrete research questions from music history and bibliography.

Acknowledgments

This work was supported by the Italian Ministry of University and Research (MUR) under the MUR Excellence Project “Inclusive Humanities: Perspectives for Development in the Research and Teaching of Foreign Languages and Literatures” of the Department of Foreign Languages and Literatures, University of Verona.

Declaration of use of Generative AI

During the preparation of this work, the authors used Claude in order to: Paraphrase and reword, Improve writing style, Grammar and spelling check. GitHub Copilot was used only as a coding assistant in the development of the RFDB Curator software and its documentation, and did not contribute to the manuscript text. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

References

[1] J. Skaug, T. Aalberg, The Concept of Work in Theater Performances, Cataloging & Classification Quarterly 62 (2024) 280–301. doi:10.1080/01639374.2024.2357208.

[2] T. Aalberg, P. Riva, M. Žumer, LRMoo: object-oriented definition and mapping from the IFLA Library Reference Model, 2026. URL: https://repository.ifla.org/handle/20.500.14598/3677.2.

[3] C. Bekiari, G. Bruseker, E. Canning, M. Doerr, P. Michon, C.-E. Ore, S. Stead, A. Velios, Definition of the CIDOC Conceptual Reference Model, version 7.1.3 ed., CIDOC CRM Special Interest Group, 2024. URL: https://cidoc-crm.org/sites/default/files/cidoc_crm_version_7.1.3.pdf.

[4] J. de Berardinis, V. A. Carriero, N. Jain, N. Lazzari, A. Meroño-Peñuela, A. Poltronieri, V. Presutti, The Polifonia ontology network: Building a semantic backbone for musical heritage, in: The Semantic Web – ISWC 2023, volume 14266 of *Lecture Notes in Computer Science*, Springer, Cham, 2023, pp. 302–322. doi:10.1007/978-3-031-47243-5_17.

[5] H. Knublauch, D. Kontokostas, Shapes constraint language (SHACL), W3C Recommendation, 2017. URL: https://www.w3.org/TR/shacl/, accessed 2026-07-17.

[6] IFLA Working Group on FRBR/CRM Dialogue, Definition of FRBRoo: A conceptual model for bibliographic information in object-oriented formalism, 2015. URL: https://repository.ifla.org/handle/20.500.14598/659.

[7] V. A. Carriero, A. Gangemi, M. L. Mancinelli, L. Marinucci, A. G. Nuzzolese, V. Presutti, C. Veninata, ArCo: The italian cultural heritage knowledge graph, in: The Semantic Web – ISWC 2019, volume 11779 of *Lecture Notes in Computer Science*, Springer, Cham, 2019, pp. 36–52. doi:10.1007/978-3-030-30796-7_3.

[8] V. Bartalesi, N. Pratelli, E. Lenzi, A knowledge graph of medieval and renaissance geographical works, Semantic Web Journal (2026).

[9] P. Bonora, A. Pompilio, Corago in LOD. The debut of an Opera repository into the Linked Data arena, JLIS.it 12 (2021) 54–72. doi:10.4403/jlis.it-12699.

[10] H. Knublauch, Form generation using SHACL and DASH, DASH Data Shapes, TopQuadrant, 2022. URL: https://datashapes.org/forms.html, accessed 2026-07-24.

[11] E. Chuc, T. Bergwinkl, I. Smessaert, D. Beeke, SHACL 1.2 user interfaces, W3C First Public Working Draft, 2026. URL: https://www.w3.org/TR/shacl12-ui/, accessed 2026-09-10.

[12] A. Sommer, N. Car, pySHACL, Software release v0.26.0, Zenodo, 2024. doi:10.5281/zenodo.10958008.

[13] T. Pellissier Tanon, Oxigraph, Software release v0.3.19, Zenodo, 2023. doi:10.5281/zenodo.8261141.

[14] S. Ahmetaj, G. Konstantinidis, M. Ortiz, P. Pareti, M. Simkus, SHACL validation under graph updates, in: The Semantic Web – ISWC 2025, volume 16140 of *Lecture Notes in Computer Science*, Springer, Cham, 2025, pp. 140–157. doi:10.1007/978-3-032-09527-5_8.

[15] Z. Zacouris, J. Ke, M. Acosta, UpSHACL: Targeted constraint validation for updates over knowledge graphs, in: The Semantic Web – ISWC 2025, volume 16140 of *Lecture Notes in Computer Science*, Springer, Cham, 2025, pp. 122–139. doi:10.1007/978-3-032-09527-5_7.

[16] M. Figuera, P. D. Rohde, M.-E. Vidal, Trav-SHACL: Efficiently validating networks of SHACL constraints, in: Proceedings of the Web Conference 2021, WWW '21, Association for Computing Machinery, New York, NY, USA, 2021, pp. 3337–3348. doi:10.1145/3442381.3449877.